

Ubiquitous computing through social objects

We're approaching a supposed digital nirvana, an ubiquitous computing environment everywhere you could want it and attuned to our needs. In a sense, it's already here in the form of micro-chipped household appliances and elevators and, of course, smartphones. But the complexity of managing this scattered environment threatens to overwhelm us — imagine the maintenance issues when we have not a handful, but a houseful of computing objects. As embedded computers outnumber the people on Earth, the computers are going to have to self-manage to a greater extent. We need to find a new balance between man and machine.

Open APIs

We could start by networking devices in an open, extensible way. Up to this point, connections between devices are often inflexibly scripted by manufacturers. But real growth will come with open APIs. The author Bruce Sterling argues that future successful products will have public application programming interfaces that allow them to be repurposed by the user in ways that the manufacturer could not foresee. HTTP and HTML produced an explosion of networked computer applications, driven by users wielding the hyperlink. Object-oriented programming created an order-of-magnitude leap in developer productivity. Now we need a higher-level protocol to reduce complexity in large, dynamic networks.

Social objects

We humans already have experience with large, dynamic networks — that's what society is. It's now been packaged in online social networks. Facebook and Twitter are not perfect, but they're good models for managing growing networks of objects. A user can easily broadcast a message, and any object that understands that message might respond — useful when we can't remember the names and abilities of every device in our environment. And more importantly, the objects can talk to each other and back to us. So the next step is to make devices more social — e.g. they should announce everything they're doing, and respond to commands. We'll add

apps to this social object network to initiate and oversee tasks on our behalf. These apps would connect groups of objects that complement each others' capabilities to accomplish new tasks together. Consumer electronics devices can be social objects. The MIT Media Lab's CE2.O consortium, directed by Henry Holtzman and Michael Bove, proposes just this: "...a new generation of consumer electronics that are highly connected, seamlessly interoperable, situation-aware and radically simpler to use." And since this is an open conversation, other objects that we haven't thought of yet will also come up with new uses for that data. The creators of the Web hadn't thought of Amazon or YouTube, either.

Sustainable computing

The materials of electronic objects often go hand-in-hand with the functional life of those objects. The physical objects tend to be cheap and flimsy, and the software becomes obsolete as quickly. If we claim to care about sustainability, the software should age as gracefully as the body. This means openness and upgradability. This is where open APIs could be a competitive advantage to manufacturers. They can gain loyal users (and more of their money) by selling apps that extend their devices' capabilities. An API means that third parties can do part of the work of

meeting customer needs (more nimbly than manufacturers can themselves) and build an ecosystem that drives sales.

Richer interactions

We experience much of the world through computers, largely restricted to our sense of sight. But we've got four other senses! We're covered in a touch sensor that goes ignored by digital interfaces. The flood of information in our lives is too large and varied to be represented in patches of pixels. Physical objects provide a wealth of data that we absorb without thinking — e.g. The heft of coins in our pocket used to indicate how much we could buy, or a grandfather clock giving us information about time, not just in the chimes every quarter hour but also in the pendulum's steady progress. We need interfaces that gracefully combine the plasticity of digital bits with the subtle richness of physical attributes. Holtzman's Information Ecology research group explores just such interfaces. The Proverbial Wallet (a project I worked on) gets harder to open as you approach your monthly budget, leading to an intuitive financial sense. Kairoscope is a timepiece that negotiates your schedule and others' so that you show up at the same time as everyone else. Tastes Like Rain flavours your toothpaste to give you a tingling sense of the weather forecast while you're brushing teeth. Mantis, a sub-\$100 home fabricator, makes physical objects that represent data in ways that flat screens cannot.

We count bullet points when shopping for consumer electronics, while we value qualitative experience in most products. Chairs, shoes — they do the same thing, so they compete on emotional qualities. How do those shoes make you feel? Is that chair comfortable and does it look good in the house? Unlike shoes and chairs, we make electronic objects do many things poorly. We need to think more holistically. Strip and polish rough features, and offer what's left to the network. We'll have built a community of objects, each honed for a different purpose, a satisfying functional and emotional extension of ourselves. We'll fulfil Sterling's goal to "design for, not just for objects or for people, but for the technosocial interactions that unite people and objects." 



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